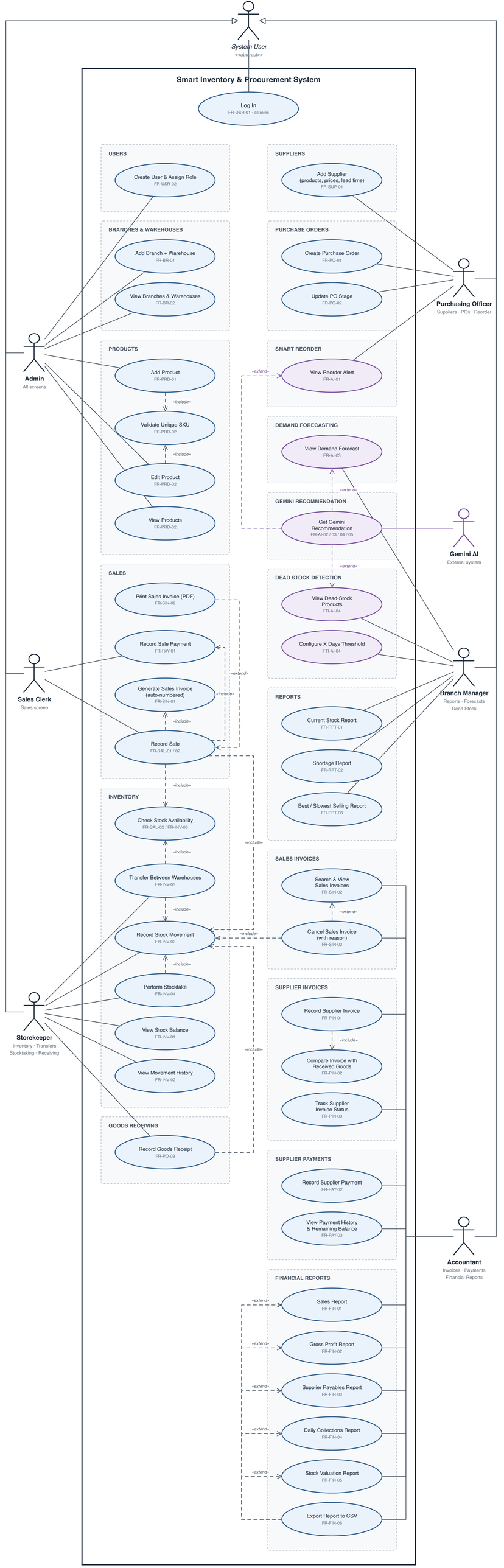
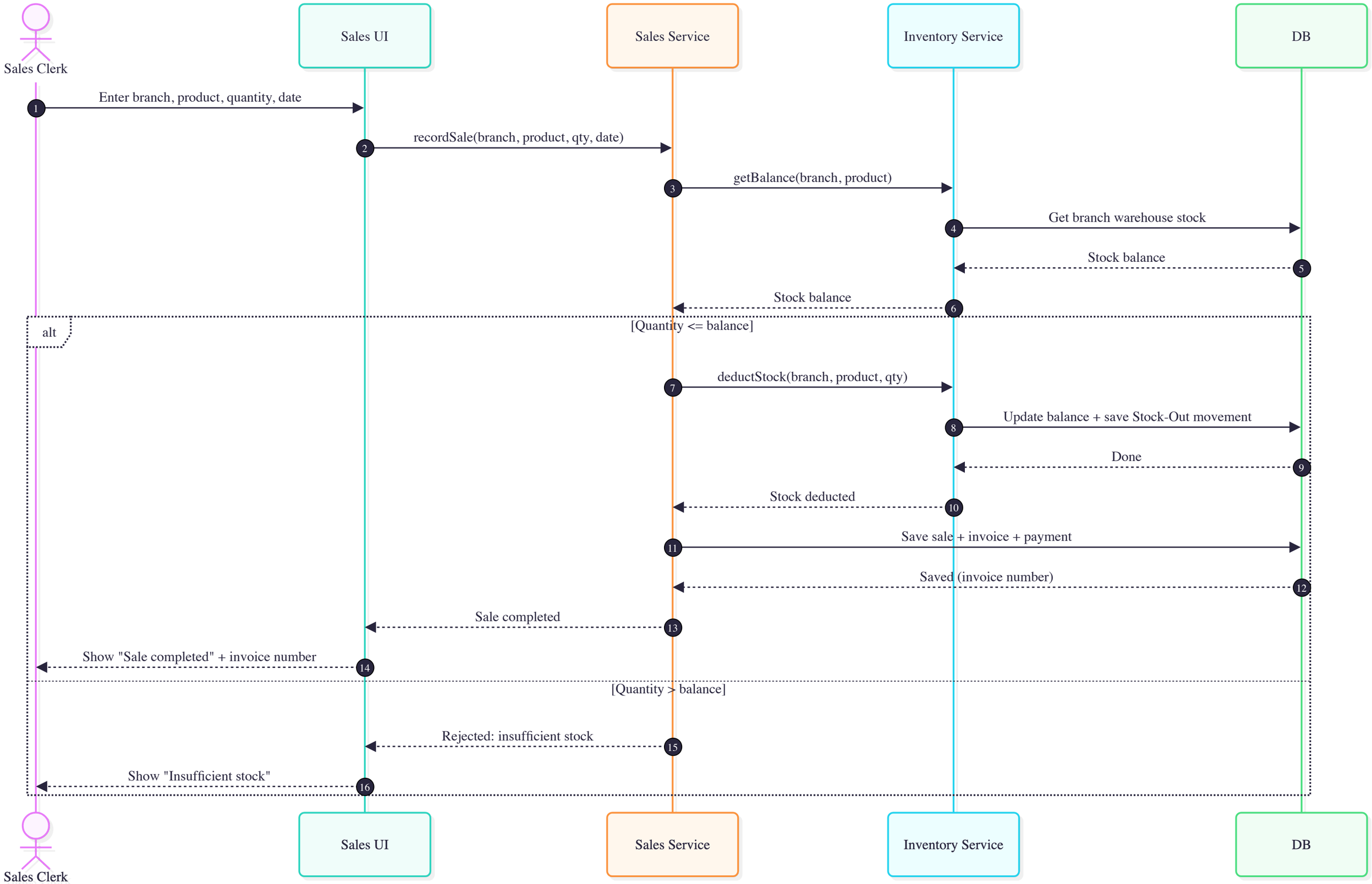
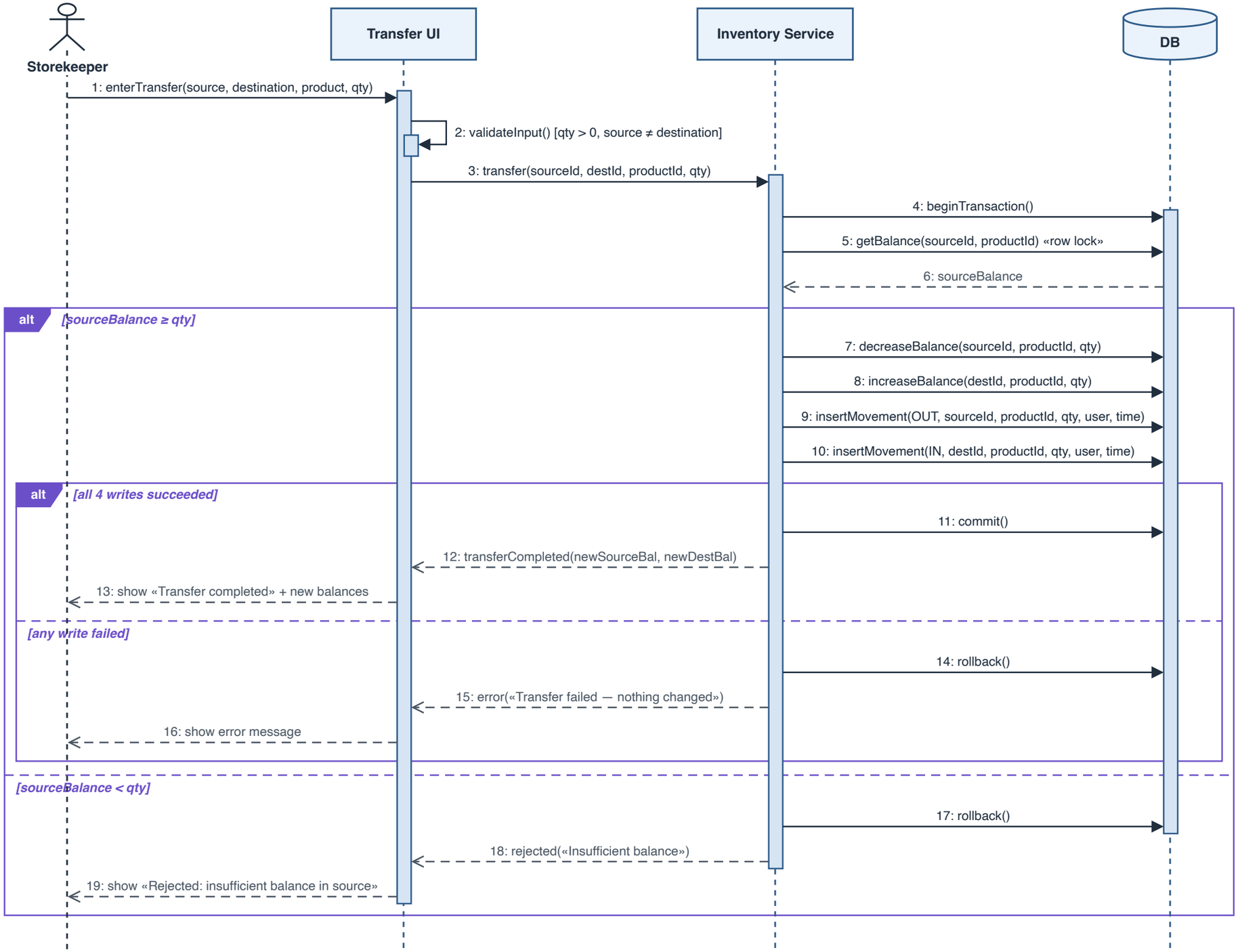

System User
<<abstract>>
Smart Inventory & Procurement System
Log In
FR-USR-01 · all roles
USERS
Create User & Assign Role
FR-USR-02
SUPPLIERS
Add Supplier
(products, prices, lead time)
FR-SUP-01
BRANCHES & WAREHOUSES
Add Branch + Warehouse
FR-BR-01
View Branches & Warehouses
FR-BR-02
PURCHASE ORDERS
Create Purchase Order
FR-PO-01
Update PO Stage
FR-PO-02
Purchasing Officer
Suppliers · POs · Reorder
Admin
All screens
PRODUCTS
Add Product
FR-PRD-01
«include»
Validate Unique SKU
FR-PRD-02
«include»
Edit Product
FR-PRD-02
View Products
FR-PRD-02
«extend»
SMART REORDER
View Reorder Alert
FR-AI-01
DEMAND FORECASTING
View Demand Forecast
FR-AI-03
«extend»
GEMINI RECOMMENDATION
Get Gemini
Recommendation
FR-AI-02 / 03 / 04 / 05
Gemini AI
External system
«extend»
DEAD STOCK DETECTION
View Dead-Stock
Products
FR-AI-04
Configure X Days Threshold
FR-AI-04
SALES
Print Sales Invoice (PDF)
FR-SIN-02
Record Sale Payment
FR-PAY-01
«extend»
Generate Sales Invoice
(auto-numbered)
FR-SIN-01
«include»
«include»
Record Sale
FR-SAL-01 / 02
Sales Clerk
Sales screen
Branch Manager
Reports · Forecasts
Dead Stock
REPORTS
Current Stock Report
FR-RPT-01
Shortage Report
FR-RPT-02
Best / Slowest Selling Report
FR-RPT-03
«include»
INVENTORY
Check Stock Availability
FR-SAL-02 / FR-INV-03
«include»
Transfer Between Warehouses
FR-INV-03
«include»
Record Stock Movement
FR-INV-02
«include»
Perform Stocktake
FR-INV-04
View Stock Balance
FR-INV-01
View Movement History
FR-INV-02
«include»
SALES INVOICES
Search & View
Sales Invoices
FR-SIN-02
«extend»
Cancel Sales Invoice
(with reason)
FR-SIN-03
«include»
SUPPLIER INVOICES
Record Supplier Invoice
FR-PIN-01
«include»
Compare Invoice with
Received Goods
FR-PIN-02
Track Supplier
Invoice Status
FR-PIN-03
«include»
GOODS RECEIVING
Record Goods Receipt
FR-PO-03
«include»
Storekeeper
Inventory · Transfers
Stocktaking · Receiving
SUPPLIER PAYMENTS
Record Supplier Payment
FR-PAY-02
View Payment History
& Remaining Balance
FR-PAY-03
Accountant
Invoices · Payments
Financial Reports
FINANCIAL REPORTS
«extend»
Sales Report
FR-FIN-01
«extend»
Gross Profit Report
FR-FIN-02
«extend»
Supplier Payables Report
FR-FIN-03
«extend»
Daily Collections Report
FR-FIN-04
«extend»
Stock Valuation Report
FR-FIN-05
Export Report to CSV
FR-FIN-06

Record Sale


Sales Clerk
Sales UI
Sales Service
Inventory Service
DB
1 Enter branch, product, quantity, date
2 recordSale(branch, product, qty, date)
3 getBalance(branch, product)
4 Get branch warehouse stock
5 Stock balance
6 Stock balance
alt
[Quantity <= balance]
7 deductStock(branch, product, qty)
8 Update balance + save Stock-Out movement
9 Done
10 Stock deducted
11 Save sale + invoice + payment
12 Saved (invoice number)
13 Sale completed
14 Show "Sale completed" + invoice number
[Quantity > balance]
15 Rejected: insufficient stock
16 Show "Insufficient stock"
Sales Clerk
Sales UI
Sales Service
Inventory Service
DB

Transfer Between Warehouses



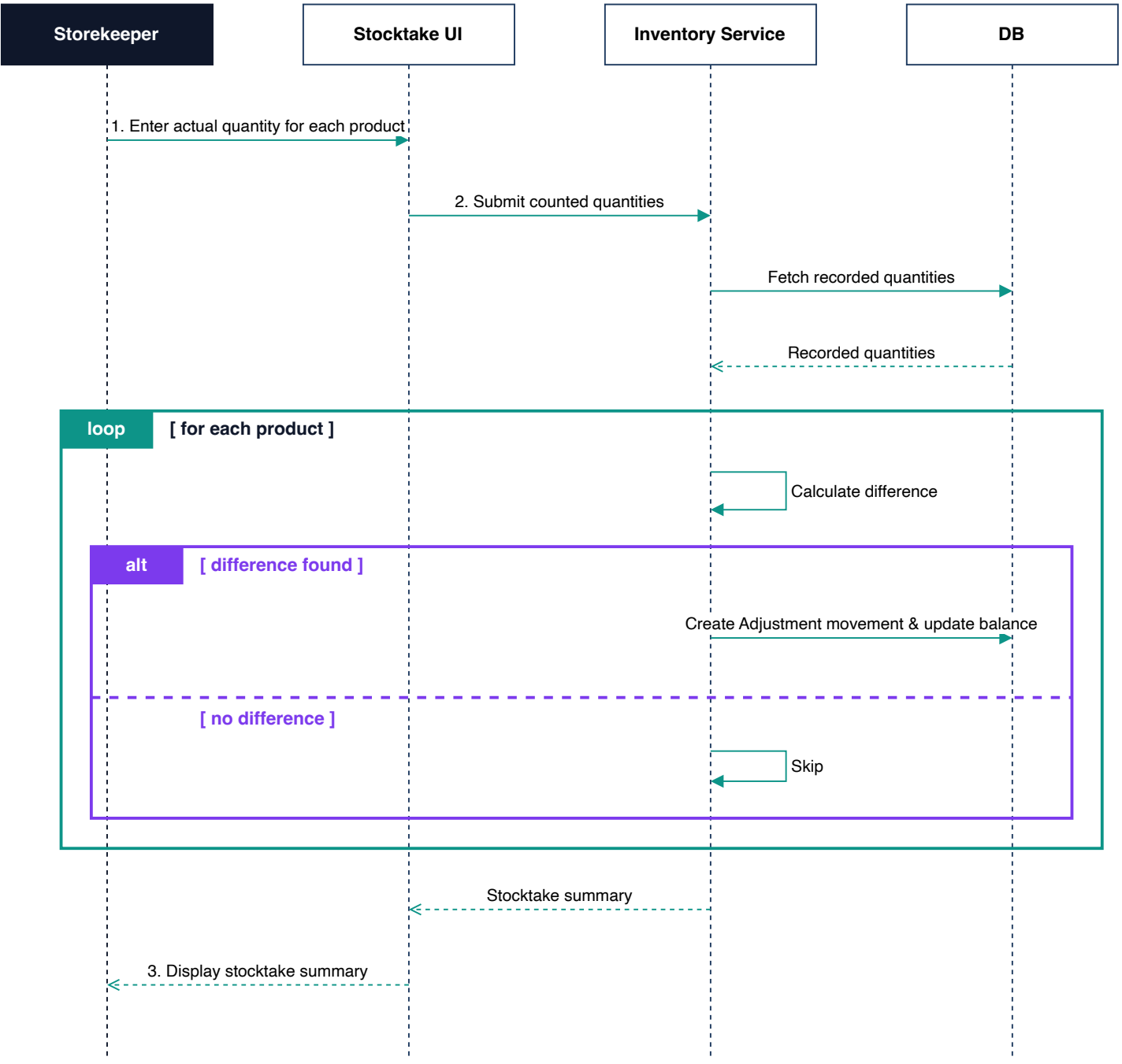
Steps 4–6 · lock first
The source balance is read inside the transaction with a row lock, so a sale or another transfer at the same moment cannot spend the same stock (NFR-REL-02).

Steps 7–10 · two movements
One OUT movement for the source and one IN movement for the destination, each stamped with user and time (FR-INV-02, NFR-SEC-05).

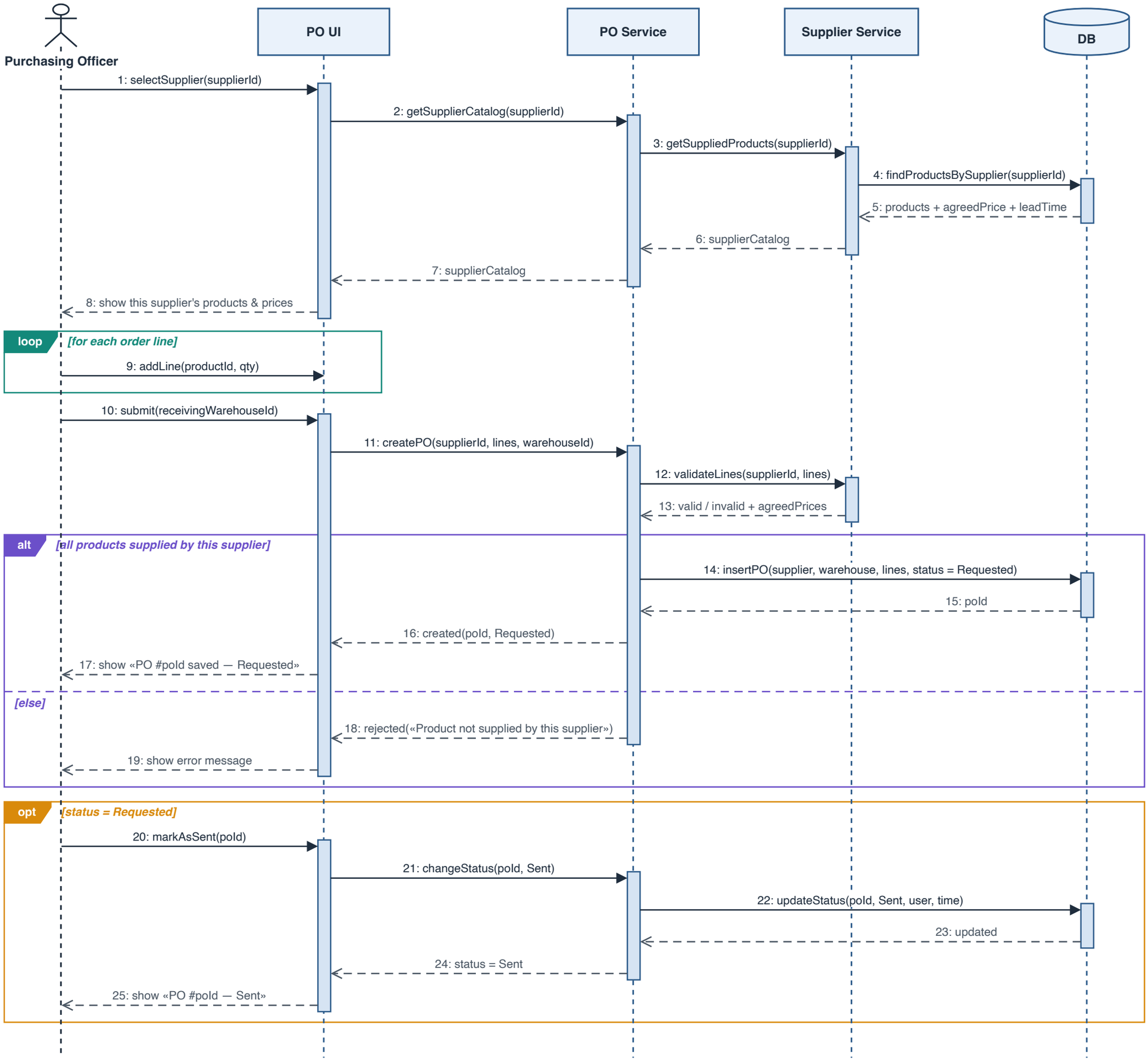
Key point · atomic transfer
All four writes are one transaction: if any write fails, rollback() undoes everything, so the source is never deducted without the destination being credited (NFR-REL-01).

Legend call (synchronous) · return / reply · activation (busy) · alt alternatives (if / else)

Perform Stocktake



Create Purchase Order



Steps 1–8 · supplier's catalog
The screen lists only the products this supplier supplies, with the agreed price and lead time (FR-SUP-01).

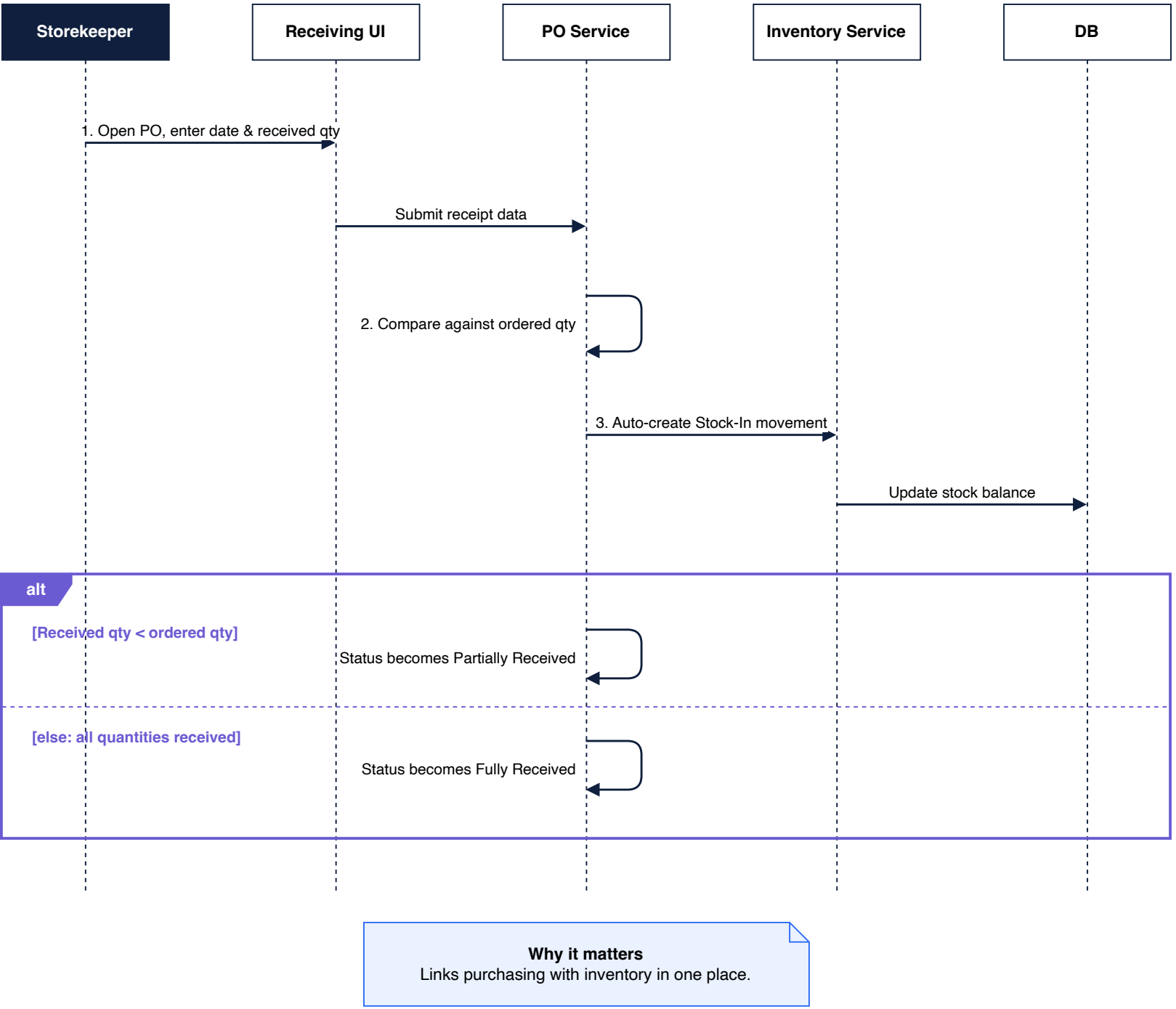
Key point · restricted products
PO Service re-checks every line against the same supplier catalog; the UI filter alone is not trusted (NFR-SEC-03).

Step 14 · price snapshot
Each PO line keeps the agreed price; supplier invoice matching uses it later (FR-PIN-02).

Steps 20–25 · status flow
When the order is sent to the supplier: Requested → Sent. Partially / Fully Received is set later by Goods Receipt, UC 5 (FR-PO-02).

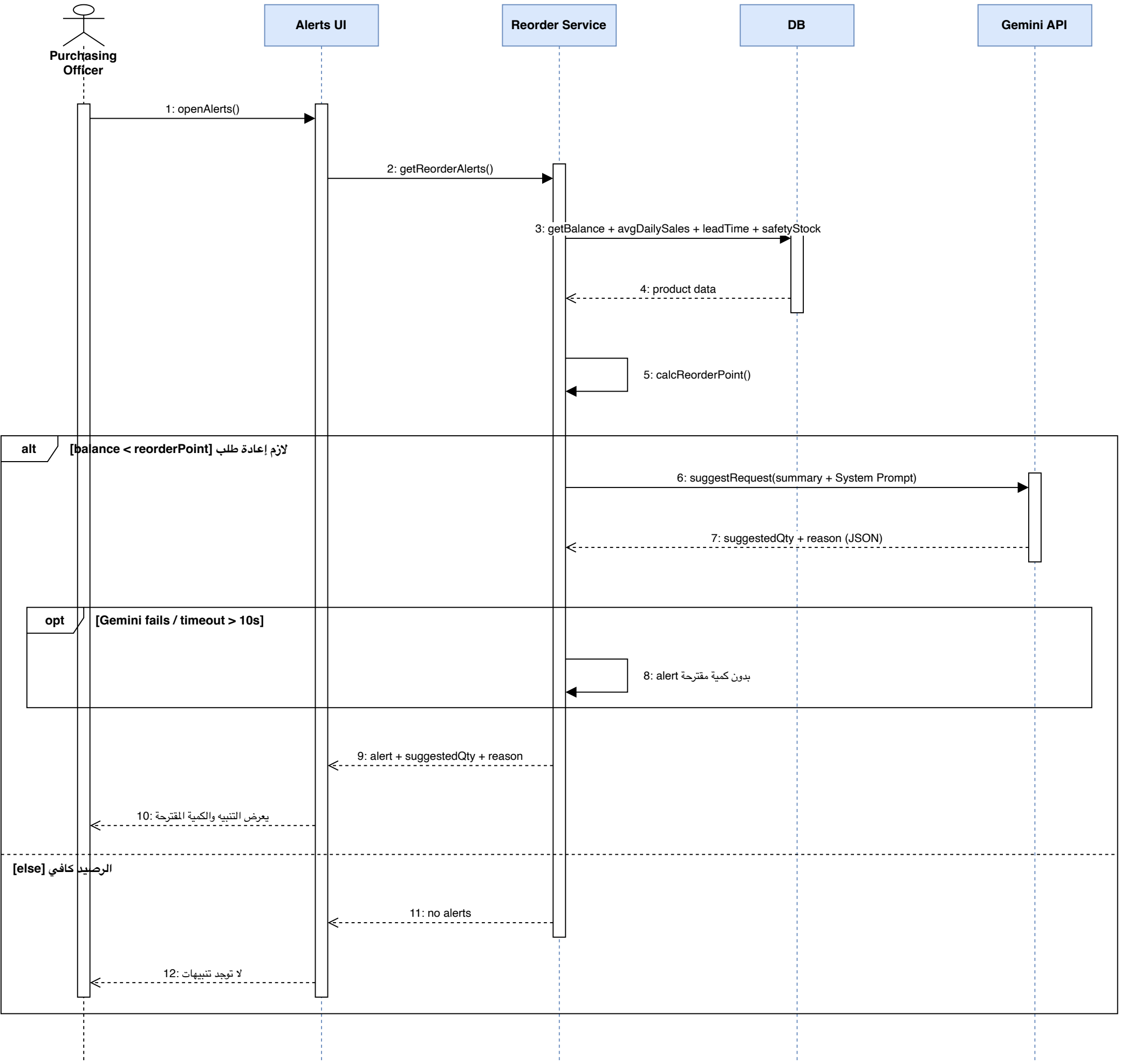
Legend call (synchronous) · return / reply · activation (busy) · alt alternatives · loop repeat · opt optional

Record Goods Receipt



Why it matters
Links purchasing with inventory in one place.

Reorder Alert & Suggested

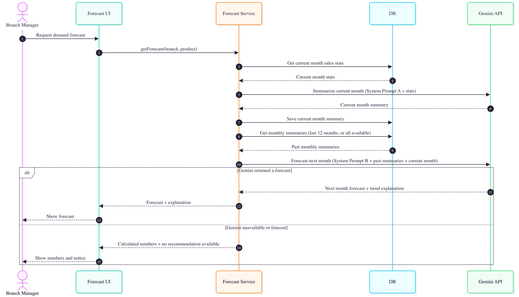


Reorder Point = (AvgDailySales × LeadTime) + SafetyStock — بيتحسب في السيستم مش في Gemini (AI-05)

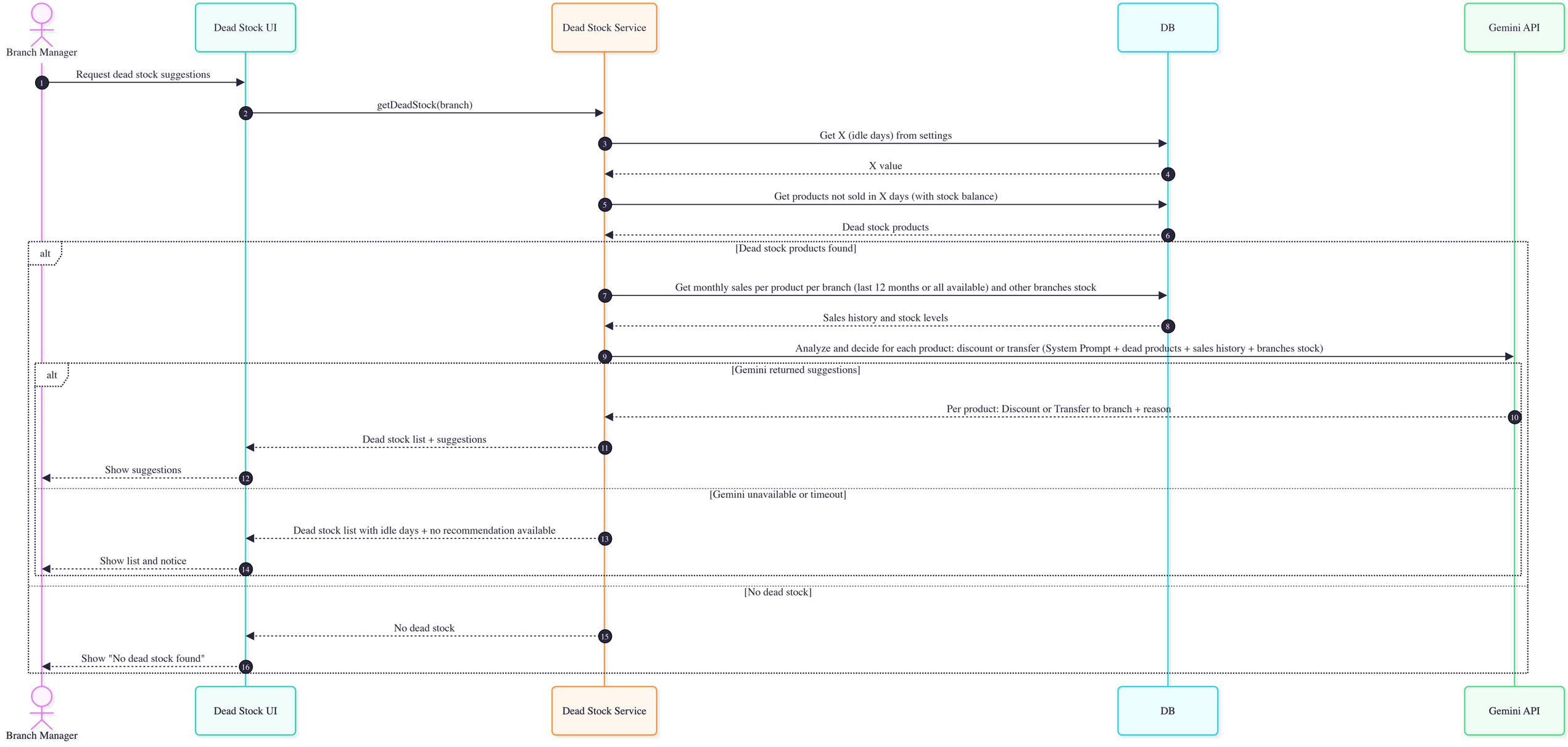
(AI-NFR-04) بيانات المنتج فقط، مفيش بيانات شخصية

وقع Gemini التنبيه يظهر حتى لو :AI-NFR-02

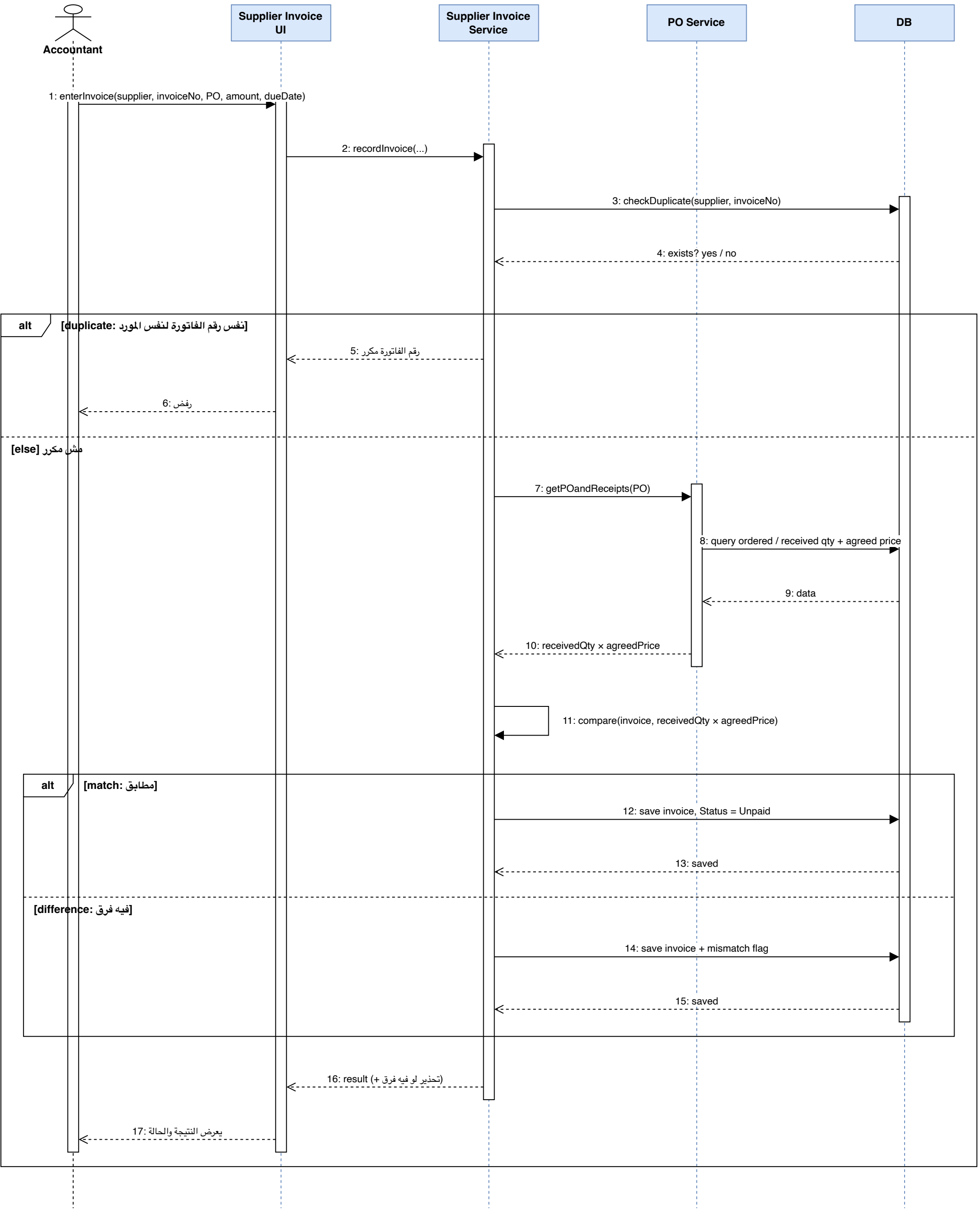
View Demand Forecast



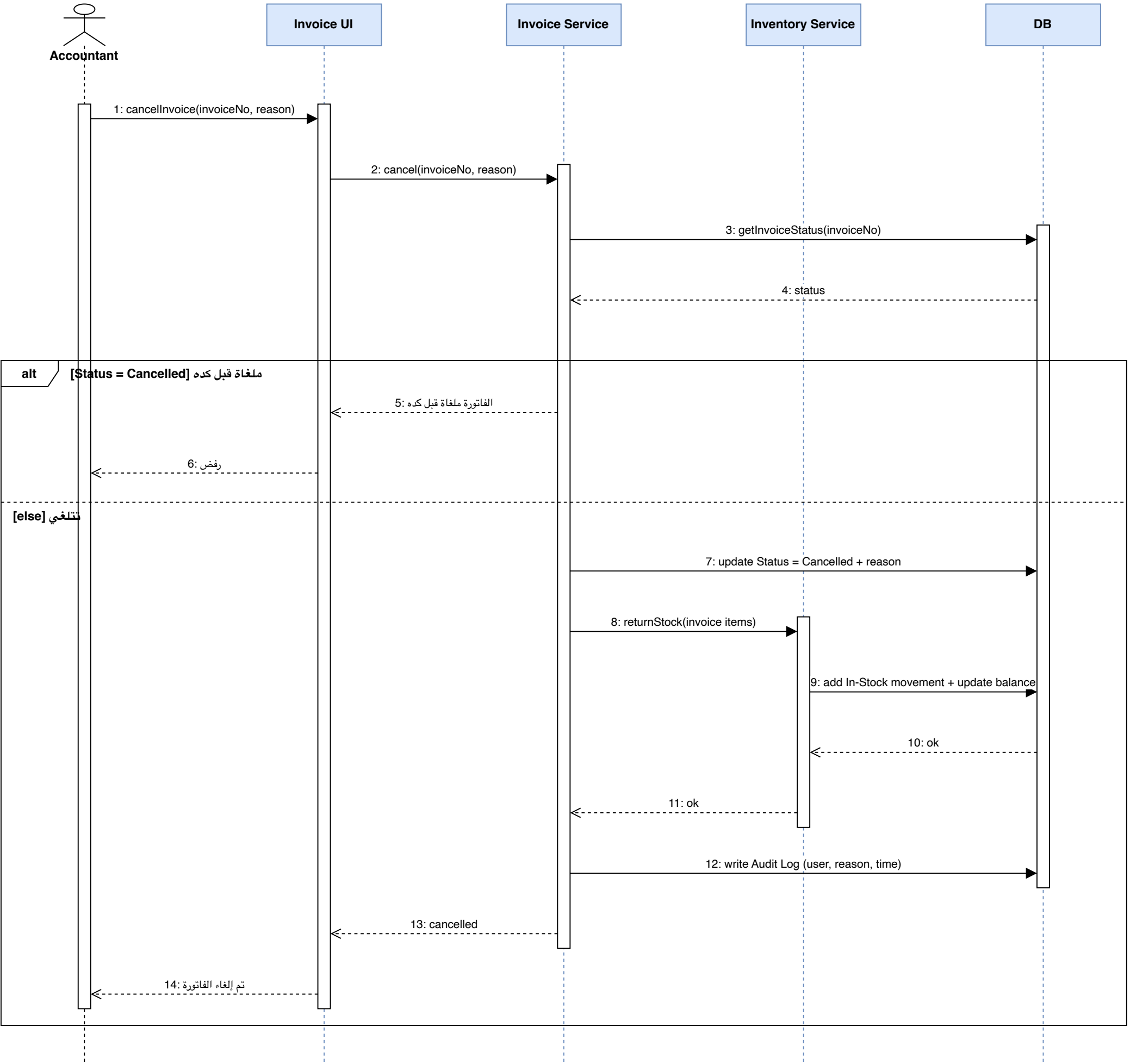
View Dead-Stock Suggestions



Supplier Invoice



Cancel Sales Invoice



الفاتورة بتتلغي بسبب ومش مفيش حذف (SEC-07): بتتمسح

Atomic Transaction: الإلغاء + رجوع المخزون + Audit Log كلهم أو لا شيء